

Collaborative Cocaine Treatment Study: **Cross-Sectional Analyses of Treatment Effects for Addiction Severity Index Legal Subscale**

Presented by Christine Killion

Source: *Psychosocial Treatments for Cocaine Dependence: National Institute on Drug Abuse Collaborative Cocaine Treatment Study*

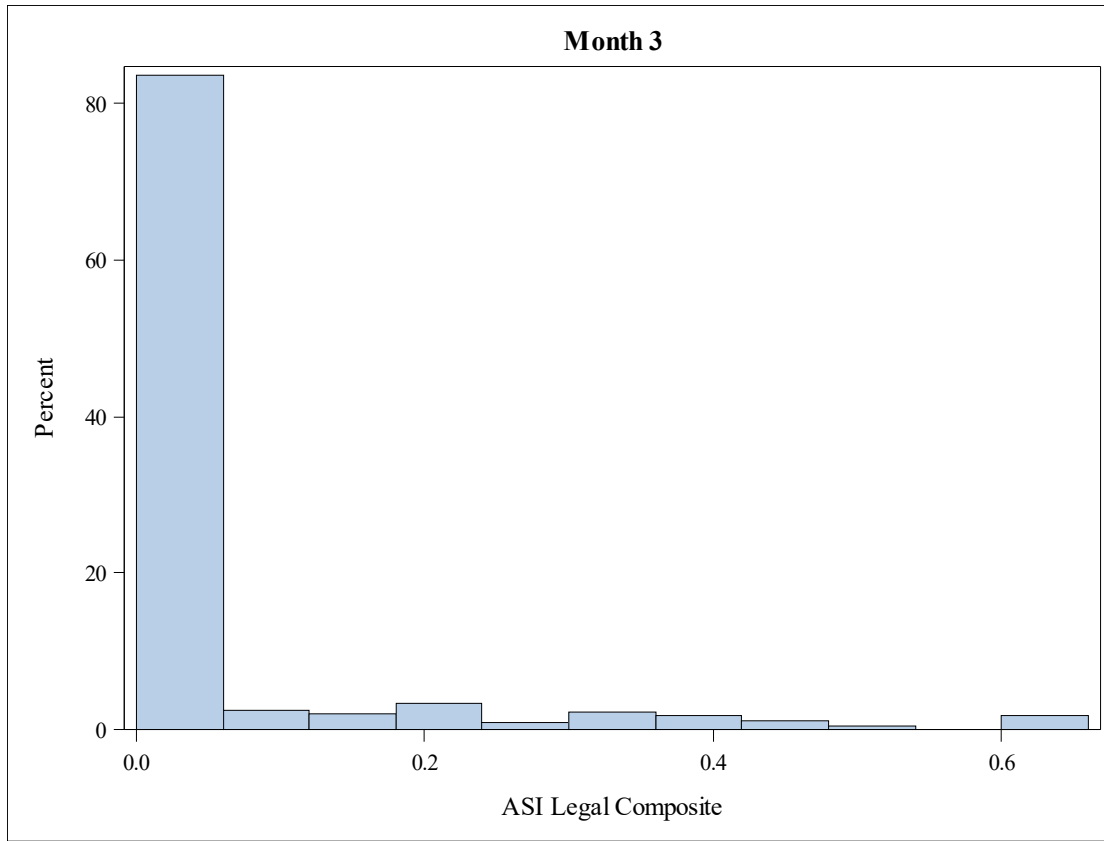
Background

- The **NIDA Collaborative Cocaine Treatment Study** randomly assigned one of four psychosocial treatments to 487 cocaine-dependent patients across five treatment centers. The active treatment period spanned six months.
- The **Addiction Severity Index (ASI)** is among the most widely used tools for assessing problems associated with substance abuse disorder.*
- The **Legal Subscale** of the ASI measures a patient's involvement with the legal system on a scale of 0 to 1, with 0 representing no reported legal problems and 1 representing severe legal problems.

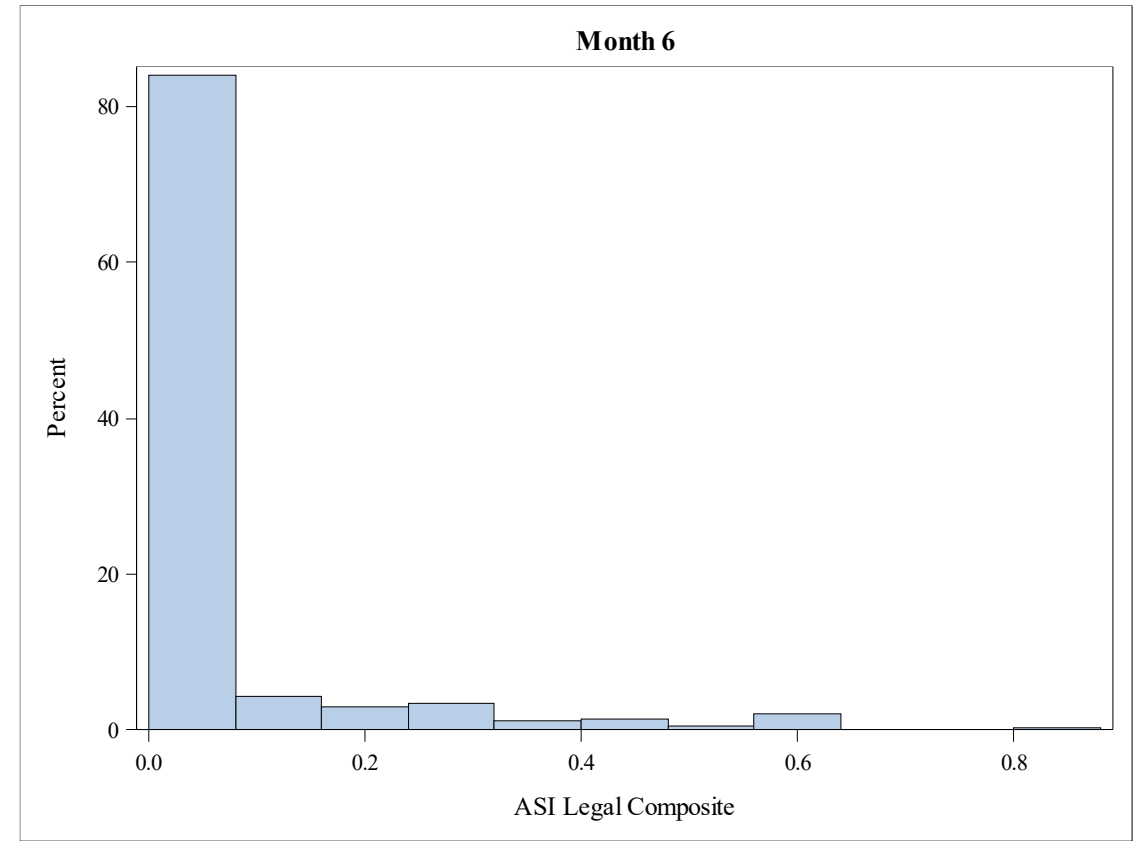
**Source: J Subst Abuse Treat. 2013 Jul 23;45(5):444–450. doi: [10.1016/j.jsat.2013.06.003](https://doi.org/10.1016/j.jsat.2013.06.003)*

Response Variable: Legal Subscale Score (LEG_SUB)

Distributions at Month 3 and Month 6



Month 3: 350 patients; 1 missing Legal Subscore



Month 6: 378 patients; 2 missing Legal Subscore

A Priori Variables and Covariates

INTERVENTION (TX_COND)

- Individual drug counseling plus group drug counseling (**IDC**)
- Cognitive therapy plus group drug counseling (**CT**)
- Supportive-expressive therapy plus group drug counseling (**SE**)
- Group drug counseling alone (**GDC**)

TREATMENT SITE (Site)

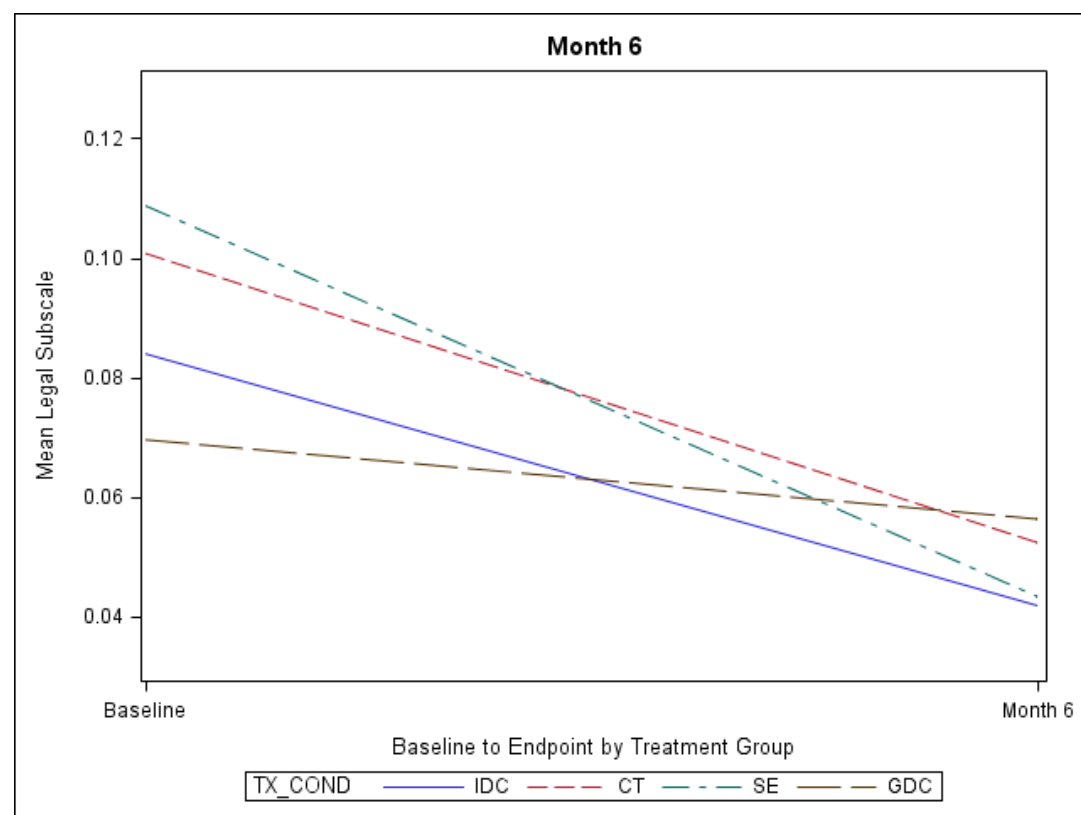
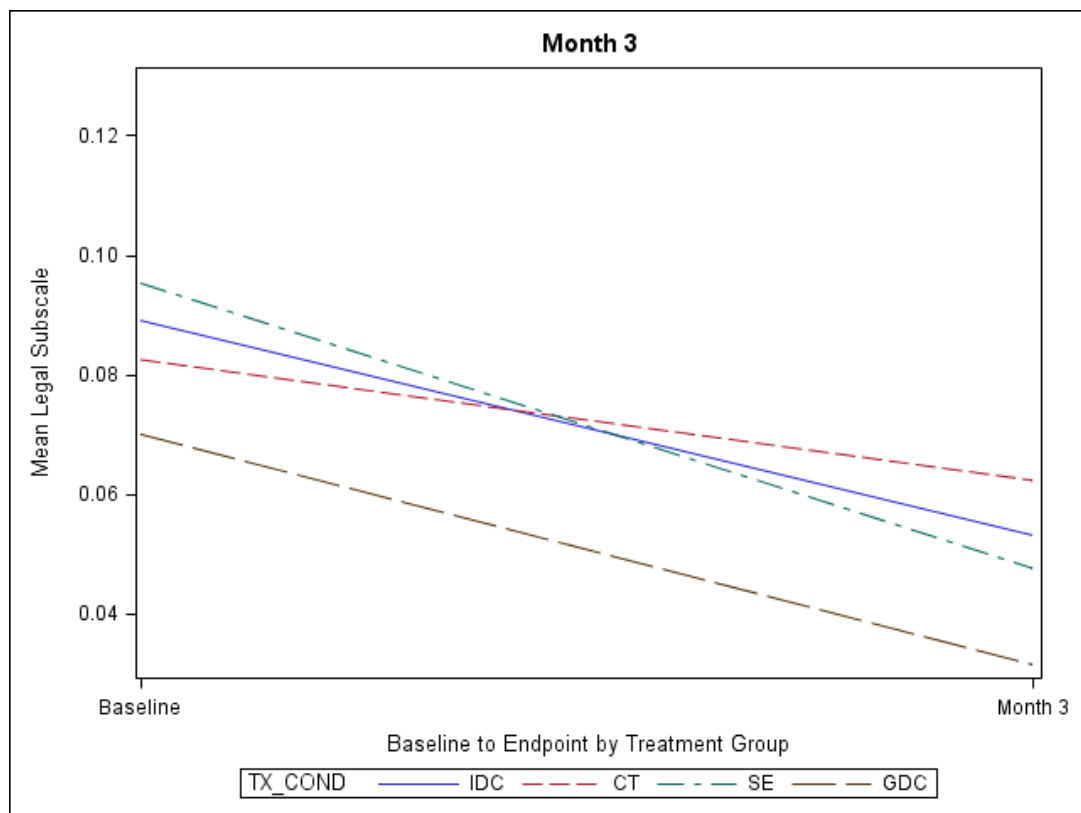
- Massachusetts General Hospital, Boston (**MGH**)
- McLean Hospital, Belmont, MA (**McLean**)
- University of Pennsylvania, Philadelphia (**Penn**)
- Western Psychiatric Institute and Clinic, Pittsburgh, PA (**WPIC**)
- Brookside Hospital, Nashua, NH (**Brookside**)

CENTERED BASELINE LEGAL COMPOSITE (CentBase)

- Baseline Legal Composite score (M0ASILE) centered around the mean baseline score: **0.0896.**

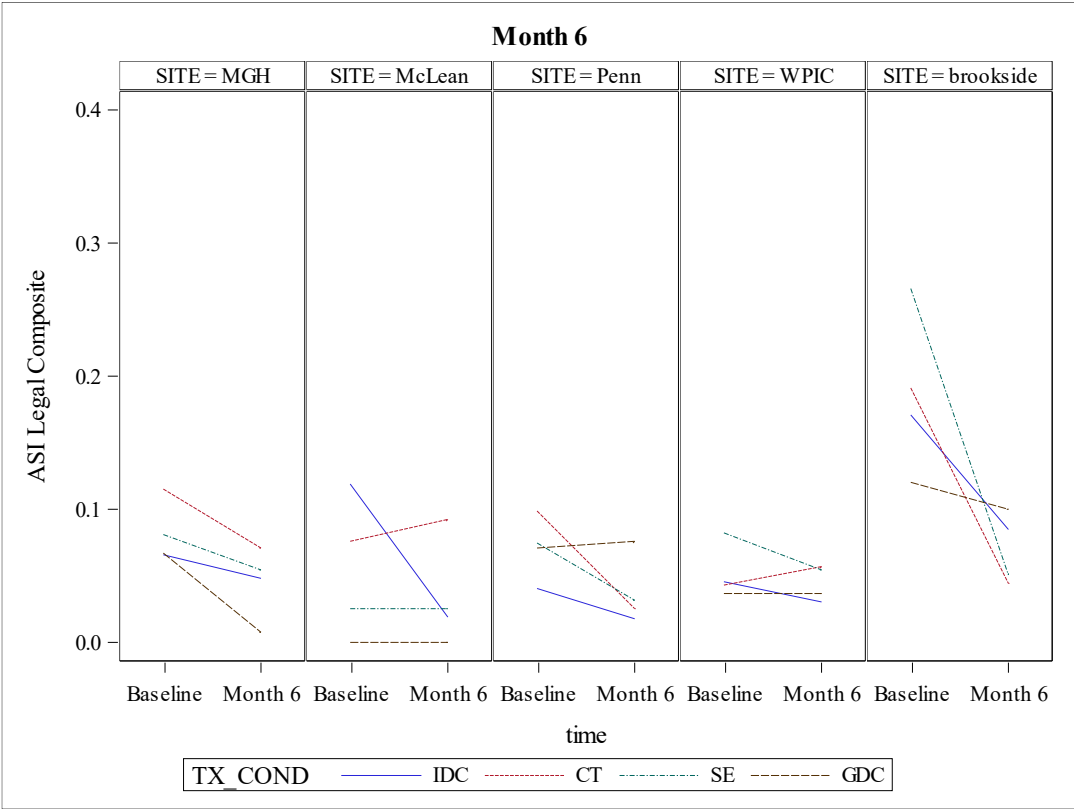
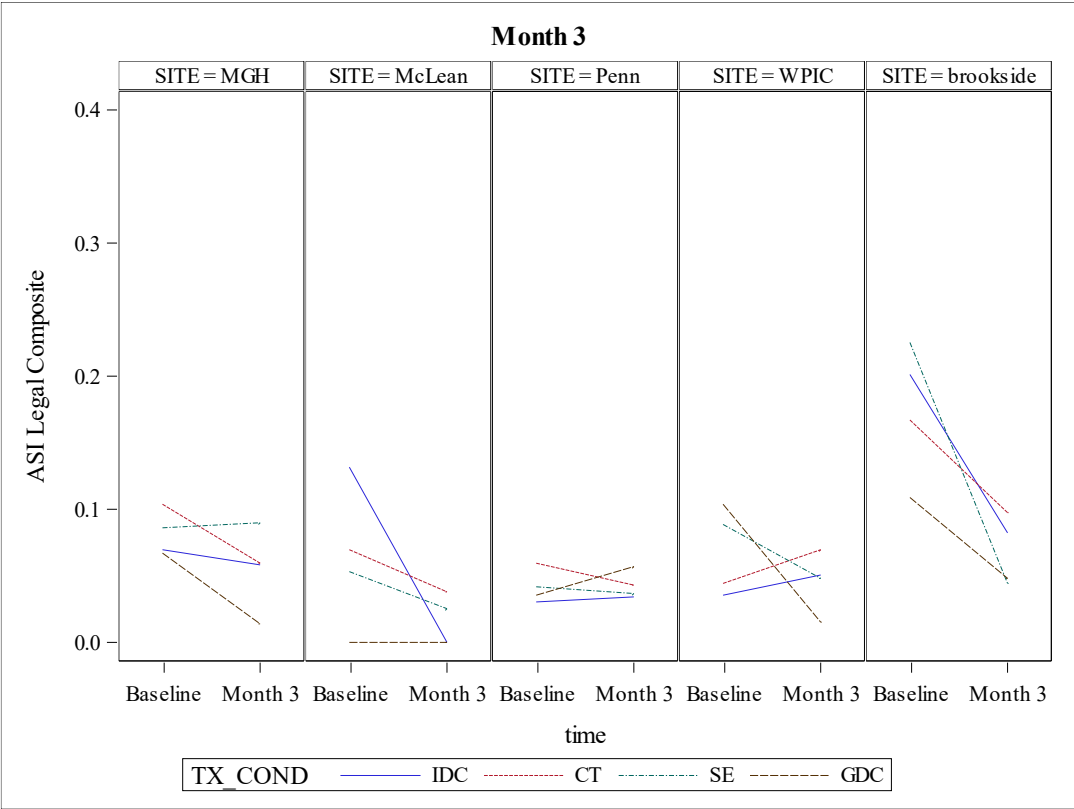
Average Legal Subscale Score

Baseline to Endpoint, by Treatment Group



Average Legal Subscale Score

Baseline to Endpoint, by Treatment Group and Site



Cross-Sectional Analyses:

Examining Effects of
Treatments, Sites, and
Baseline scores for
Legal Subscale at
Month 3 and Month 6.

Questions:

Is there a treatment effect for legal problems at month 3? At month 6? Is there a difference?

Is there a difference in legal problems across interventions when considering starting baseline scores?

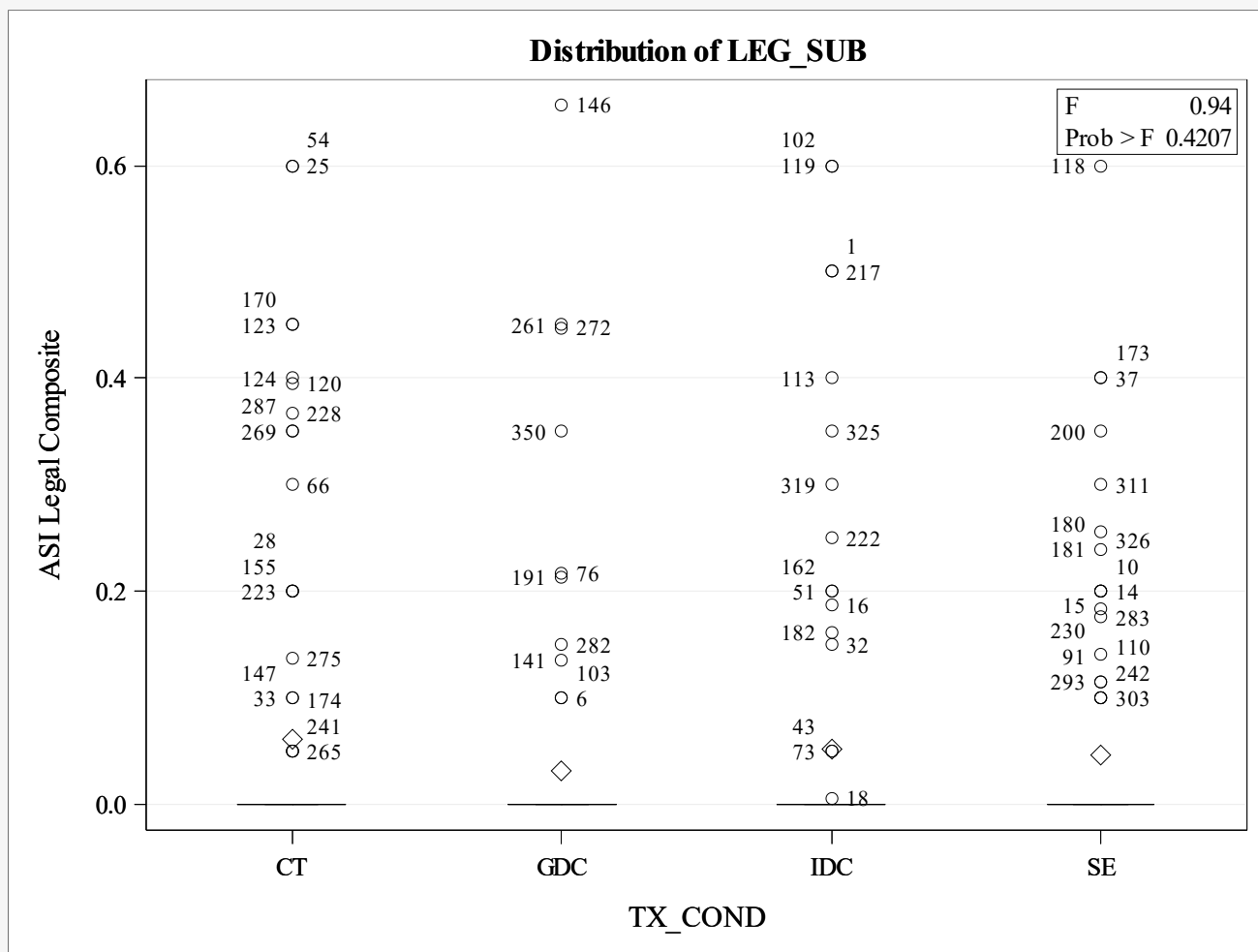
Do the differences in legal problems vary across interventions and treatment sites when considering starting baseline scores?

Models

The following cross-sectional analyses use analysis of variance (ANOVA) and analysis of covariance (ANCOVA) models to assess the treatment effect on Legal Subscale scores.

- The ANCOVA model includes a centered baseline covariate to adjust the intervention groups so that they have the same on-average baseline Legal Subscale scores.
- Least-Squares Means estimates (unadjusted) indicate any significant differences in average Legal Subscale scores between treatment groups and sites.

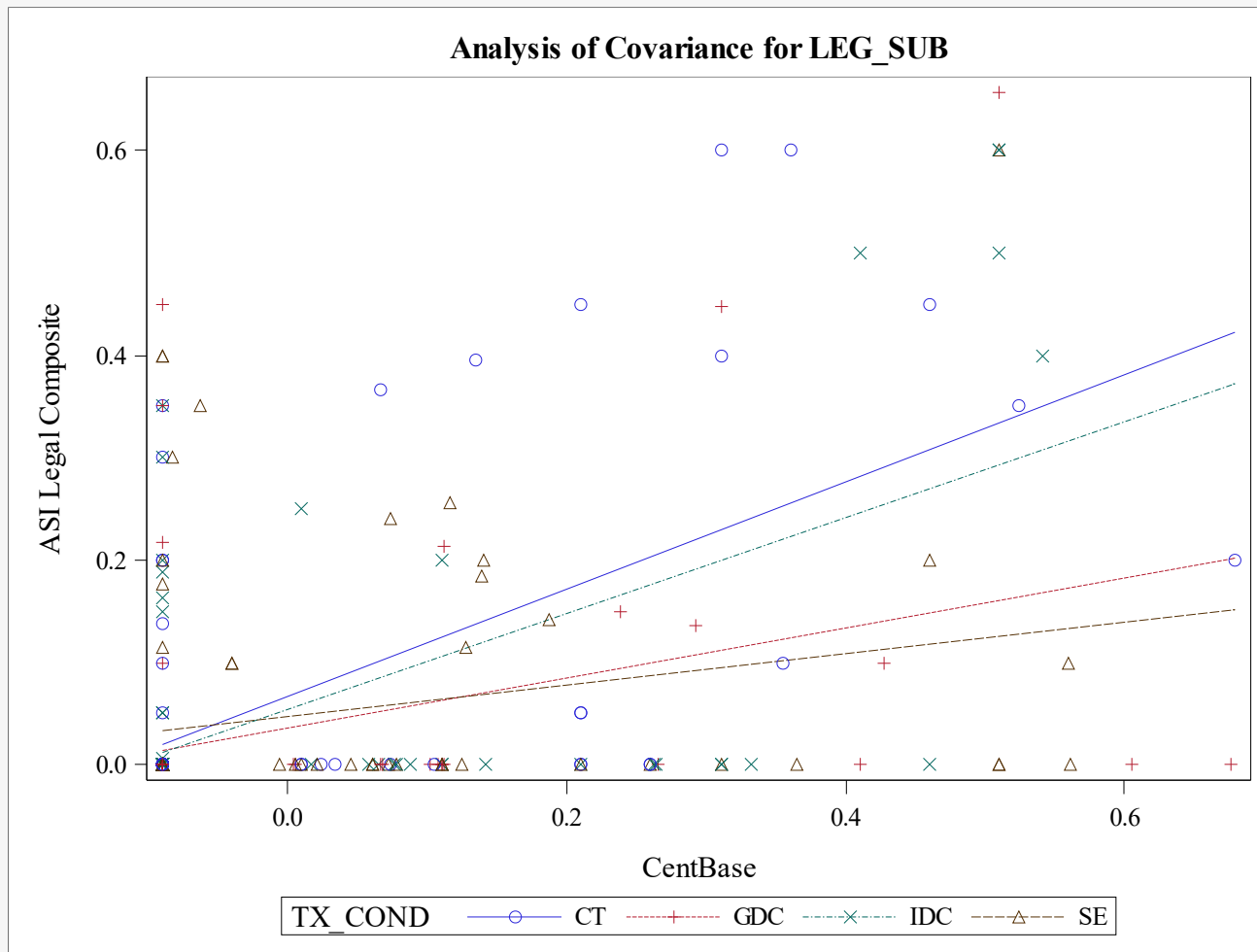
Goodness of Fit: Model assumptions tested using these tests and $\alpha=0.05$ significance level: Shapiro-Wilk, Levene's, Durbin-Watson.



Month 3: One-Way ANOVA Modeling Treatment on Legal Subscale

Without considering any covariates, there does not appear to be an overall treatment effect for Legal Subscale scores in Month 3 ($p=0.4207$).

This model explains less than 1% of the variance in Legal Subscale.

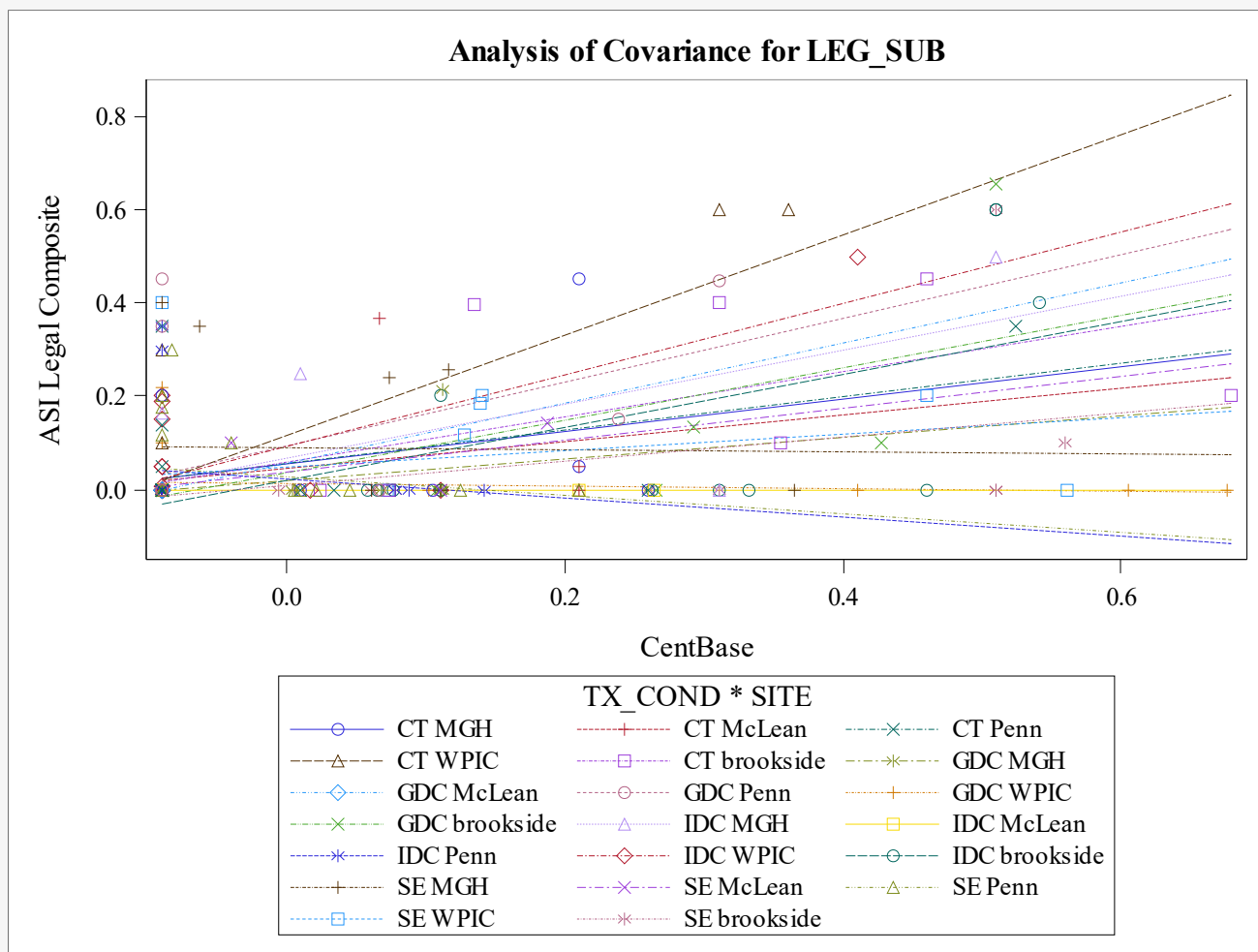


Month 3: One-Way ANCOVA Modeling Interaction of Treatment and Centered Baseline

At least one of the variables is significantly related to Legal Subscale scores at Month 3 ($p < 0.0001$), and it is the covariate: centered baseline.

This model explains 26.2% of the variance in Legal Subscale.

Violation of equality of slopes:
The interaction between treatment and centered baseline is significant ($p = 0.0003$).



Month 3:

Two-Way ANCOVA Modeling
Three-Way Interaction of
Treatment, Site, and
Centered Baseline

At least one of the variables is
significantly related to Legal
Subscale scores at Month 3
($p < 0.0001$).

This model explains 38.6% of the
variance in Legal Subscale.

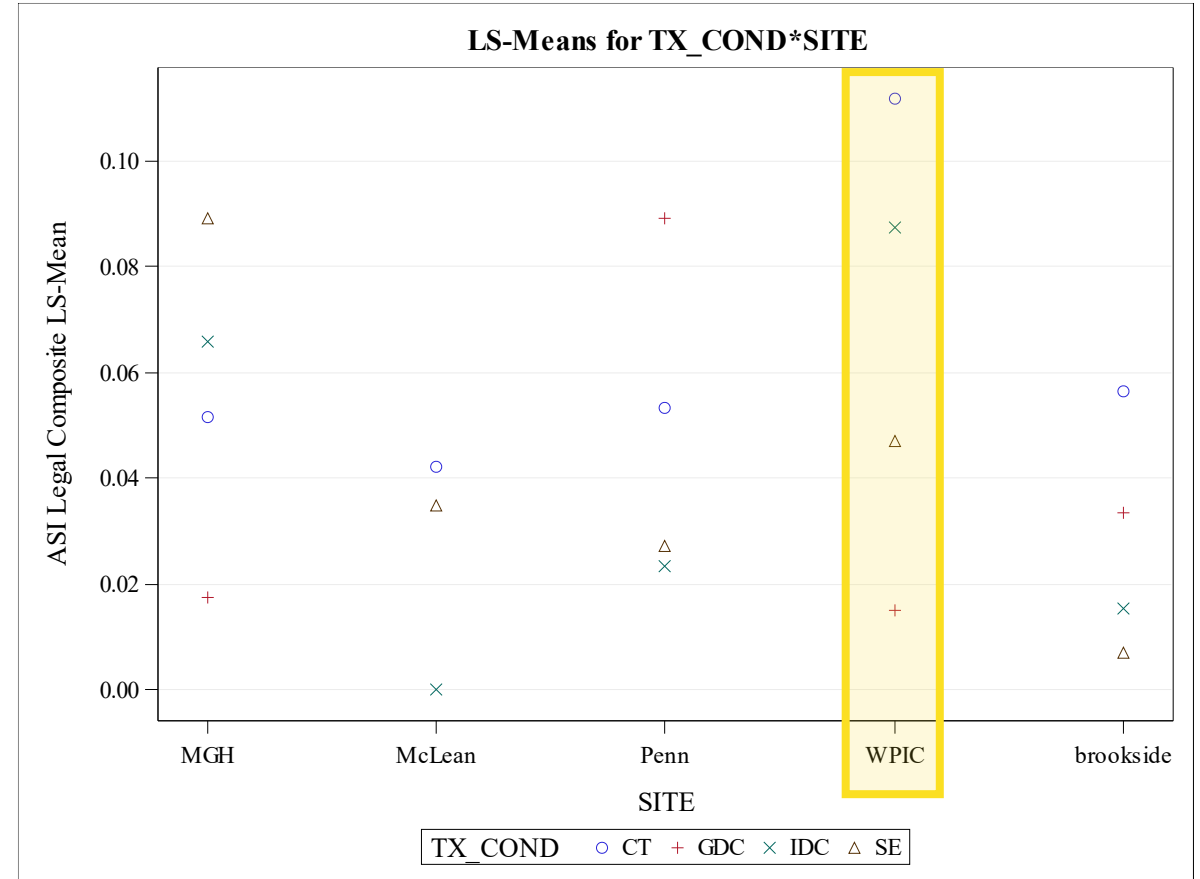
There is a **significant three-way
interaction** between treatment,
centered baseline, and site
($p < 0.0001$).

Month 3 Results

Legal Subscale scores varied differentially on average across Intervention and Site when moderating for baseline scores.

Driving this three-way interaction are differences in mean scores across treatment groups at **WPIC**.

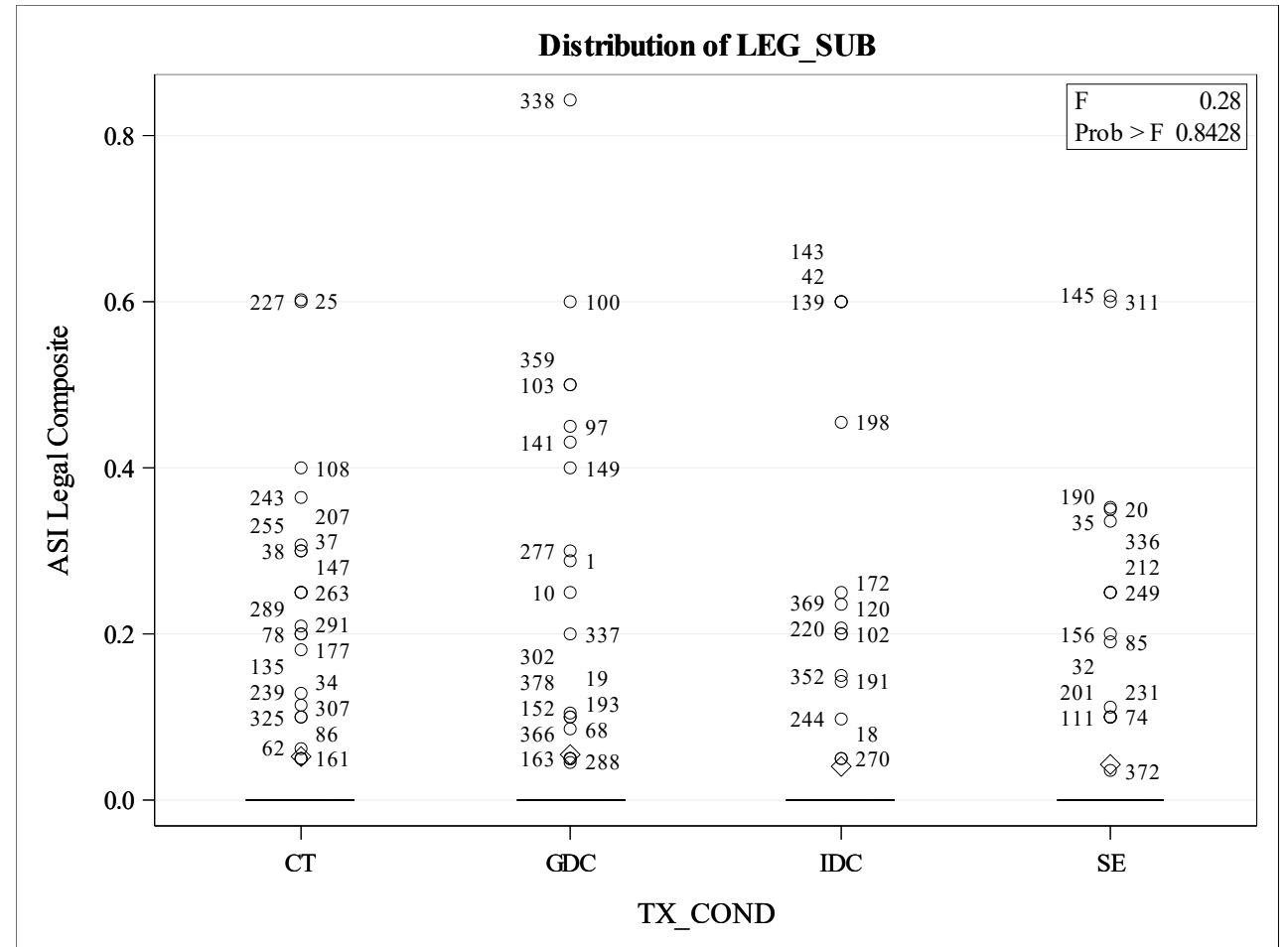
- The estimated average GDC Legal Subscale score at Month 3 (0.0149) is significantly lower than the estimated average scores for CT (0.112) and IDC (0.0872).
- The average SE score (0.047) is significantly lower than that of CT.



Month 6: One-Way ANOVA Modeling Treatment on Legal Subscale

Without considering any covariates, there does not appear to be a significant difference in mean Legal Subscale scores across Treatment groups in Month 6 ($p=0.8428$).

This ANOVA model explains less than 1% of the variance in Legal Subscale.

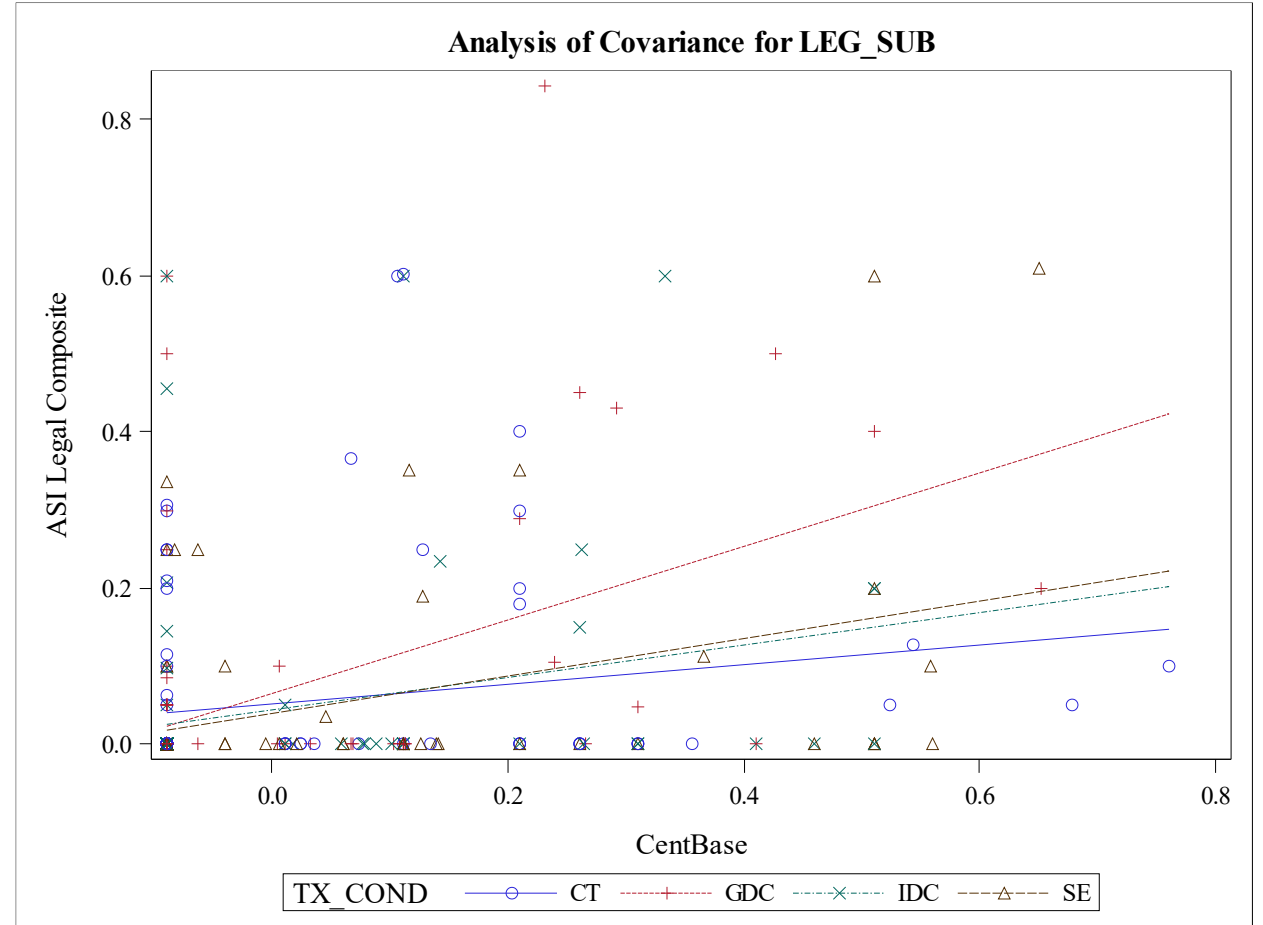


Month 6: One-Way ANCOVA Modeling Interaction of Treatment and Centered Baseline

At least one of the variables is significantly related to Legal Subscale scores at Month 6 ($p < 0.0001$), and it is the covariate: centered baseline.

This model explains 13.6% of the variance in Legal Subscale.

Violation of equality of slopes: The interaction between Treatment group and Centered Baseline is significant ($p = 0.0149$).

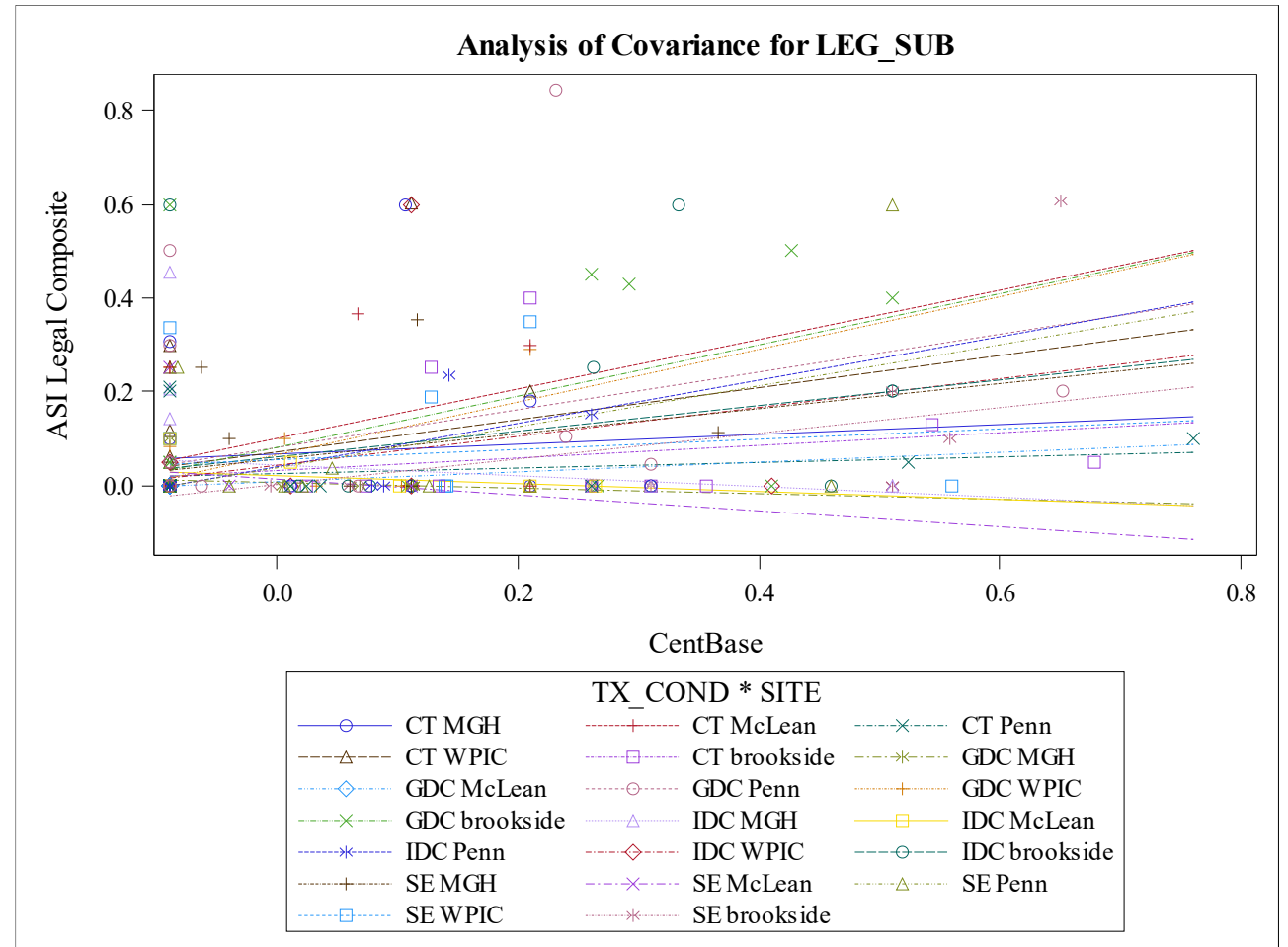


Month 6: Two-Way ANCOVA Modeling Three-Way Interaction of Treatment, Site, and Centered Baseline

At least one of the variables is significantly related to Legal Subscale scores at Month 6 ($p < 0.0002$).

This model explains 19.6% of the variance in Legal Subscale.

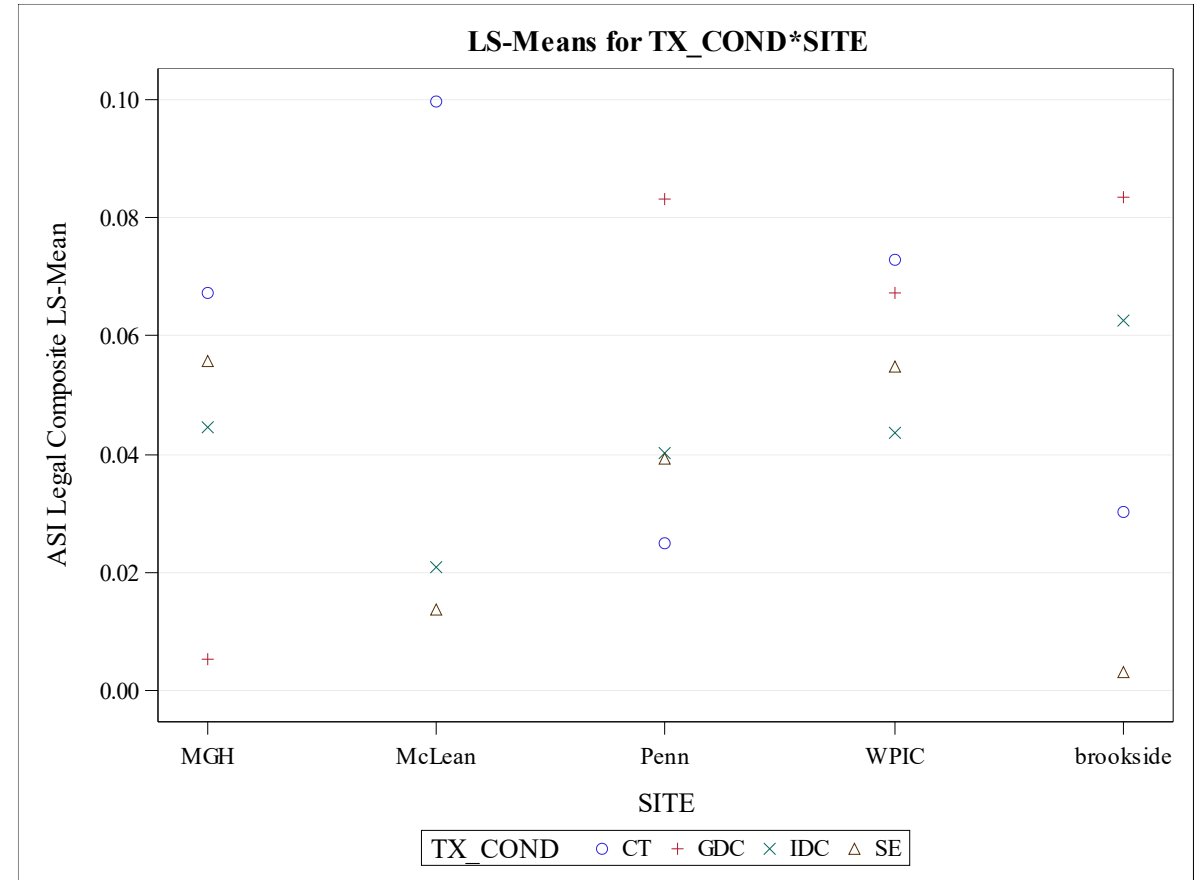
There is **no significant three-way interaction** between Treatment, Site, and Centered Baseline ($p = 0.486$).



Month 6 Results

Treatment effect on Legal Subscale varies depending on baseline scores but not depending on site.

There does not appear to be a significant main effect associated with site ($p=0.9841$).



Month 3 vs. Month 6

There is no treatment effect on the severity of legal problems at month 3, nor at month 6.

Treatment effect on Legal Subscale scores varies depending on baseline scores at both Month 3 and Month 6.

In Month 3, treatment effect on Legal Subscale scores varied differentially across sites and depending on baseline score.

However, in Month 6, treatment effect only varied depending on baseline score, and not differentially across Sites.

Conclusion

- The ASI Legal Subscale alone would not be a useful measure for assessing the effectiveness of cocaine substance abuse psychosocial interventions.
- There are likely other variables and covariates that help explain the variation in Legal Subscale scores.